



VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-1:Lecture-1

Number Systems

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Module-1 (Part-1)

- ❖ Number systems and conversions
- ❖ r 's and $(r-1)$'s compliment
- ❖ Binary signed and unsigned numbers
- ❖ Binary arithmetic operations
- ❖ Weighted and non-weighted binary codes

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Lecture-1

- ❖ Digital Systems
- ❖ Number system
- ❖ Understanding Decimal number systems
- ❖ Understanding Binary number systems
- ❖ Understanding Octal number systems
- ❖ Understanding Hexadecimal number systems

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Number Systems

DIGITAL SYSTEMS

- ✓ **Digital systems** are used in communication, business transactions, traffic control, spacecraft guidance, medical treatment, weather monitoring, the Internet, digital telephones, digital televisions, digital cameras, handheld devices, and, of course, digital computers.
- ✓ Digital computers follow a **sequence of instructions**, called a **program**, that operates on given data.
- ✓ One characteristic of **digital systems** is their ability to represent and manipulate **discrete elements of information**.
- ✓ Examples: **10 decimal digits,**
26 letters of the alphabet,
52 playing cards,
64 squares of a chessboard, etc.

DIGITAL SYSTEMS

- ✓ These discrete elements were the digits. From this application, the term digital computer emerged. **Discrete elements of information** are represented in a digital system by **physical quantities called signals** (Electrical signals such as **voltage and current**).
- ✓ The signals is represented either as **0 or 1, i.e., binary values**.
- ✓ Thus, these discrete elements of information are represented with **groups of bits** called **binary codes**.
- ✓ For example, the decimal digits 0 through 9 are represented in a digital system with a code of four bits

e.g.: **number 7 – 0111**

number 9 – 1001

NUMBER SYSTEMS

- ✓ The **representation** of a number system is given by

$$(N)_b \longrightarrow \begin{array}{l} N \text{ is the number} \\ b \text{ is the radix or base} \end{array}$$

- ✓ The **characteristics** of a number system
 1. The base is equal to the number of digits
 2. The largest digit is one less than the base
 3. Each digit is multiplied by the base raised to an appropriate power depending upon the digit position to get its place value.

NUMBER SYSTEMS

- ✓ In general, digit positions can be represented as

$$(a_n a_{n-1} \dots a_1 a_0 . a_{-1} a_{-2} \dots a_{-(m-1)} a_{-m})$$

- ✓ The number can be represented as

$$(N)_b = a_n \times b^n + a_{n-1} \times b^{n-1} + \dots + a_0 \times b^0 + a_{-1} \times b^{-1} + a_{-2} \times b^{-2} + \dots + a_{-m} \times b^{-m}$$

a_n $\longrightarrow$ n^{th} digit

N $\longrightarrow$ The entire number

b $\longrightarrow$ Base

UNDERSTANDING DECIMAL NUMBERS

- ✓ Decimal number system has ten separate digits: (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
- ✓ It is represented as base 10

$$(N)_{10}$$

- ✓ Digit Positions:

5	6	7	8	.	3	4
10^3	10^2	10^1	10^0		10^{-1}	10^{-2}

- ✓ Examples:

$$(8653)_{10} = 8 \times 10^3 + 6 \times 10^2 + 5 \times 10^1 + 3 \times 10^0$$

$$(97654.35)_{10} = 9 \times 10^4 + 7 \times 10^3 + 6 \times 10^2 + 5 \times 10^1 + 4 \times 10^0 + 3 \times 10^{-1} + 5 \times 10^{-2}$$

UNDERSTANDING BINARY NUMBERS

- ✓ Binary number system has two separate digits: (0 and 1)
- ✓ It is represented as base 2

$$(N)_2$$

- ✓ Digit Positions:

1	0	1	1	.	0	1	1
2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}

- ✓ In the modern digital systems and computers, the data operations are pursued by binary numbers
- ✓ Examples: $(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$$(100011.101)_2 = 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

UNDERSTANDING BINARY NUMBERS

- ✓ Groups of **eight bits** are called
a **byte**

$(11001001)_2$

- ✓ Groups of **four bits** are called
a **nibble**

$(1101)_2$

n	2^n
0	$2^0=1$
1	$2^1=2$
2	$2^2=4$
3	$2^3=8$
4	$2^4=16$
5	$2^5=32$
6	$2^6=64$
7	$2^7=128$

n	2^n
8	$2^8=256$
9	$2^9=512$
10	$2^{10}=1024$
11	$2^{11}=2048$
12	$2^{12}=4096$
20	$2^{20}=1M$
30	$2^{30}=1G$
40	$2^{40}=1T$

Mega

Giga

Tera

UNDERSTANDING OCTAL NUMBERS

- ✓ Octal number system has 8 separate digits: (0, 1, 2, 3, 4, 5, 6, 7)
- ✓ It is represented as base 8

$(N)_8$

- ✓ Digit Positions:

5	3	7	1	.	2	6	5
8^3	8^2	8^1	8^0		8^{-1}	8^{-2}	8^{-3}

- ✓ Examples: $(567)_8 = 5 \times 8^2 + 6 \times 8^1 + 7 \times 8^0$

$$(121.41)_8 = 1 \times 8^2 + 2 \times 8^1 + 1 \times 8^0 + 4 \times 8^{-1} + 1 \times 8^{-2}$$

UNDERSTANDING HEXADECIMAL NUMBERS

- ✓ Hexadecimal number system has 16 separate digits: (0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and A, B, C, D, E, F)
- ✓ In this, A to F represents 10 to 15.
- ✓ It is represented as base 16

$$(N)_{16}$$

- ✓ Digit Positions:

5	C	7	F	.	2	E	A
16^3	16^2	16^1	16^0		16^{-1}	16^{-2}	16^{-3}

- ✓ Hexadecimal codes are used in many microcontrollers such as 8051, AVR, etc.
- ✓ Examples: $(1DA7)_{16} = 1 \times 16^3 + 13 \times 16^2 + 10 \times 16^1 + 7 \times 16^0$

$$(23ADB.FA)_{16} = 2 \times 16^4 + 3 \times 16^3 + 10 \times 16^2 + 13 \times 16^1 + 11 \times 16^0 + 15 \times 16^{-1} + 10 \times 16^{-2}$$

NUMBER SYSTEMS: SUMMARY

System	Base	Symbols	Used by humans?	Used in computers?
Decimal	10	0, 1, ... 9	Yes	No
Binary	2	0, 1	No	Yes
Octal	8	0, 1, ... 7	No	No
Hexadecimal	16	0, 1, ... 9, A, B, C, D, E, F	No	No

Thank You